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Analysis of the correlation between plasma plume and keyhole behavior in laser metal welding for the modeling of the keyhole geometry

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Acknowledgments

Highlights

- The dynamics of keyhole and plume are well correlated, especially for high laser power and feed rate.
- The trend of keyhole and plume behavior changes when reaching a threshold laser power.
- Keyhole stability for medium laser power is predominantly determined by evaporation processes.
- With high feed rate and high laser power the determining factor of keyhole stability is the melt pool.

Abstract

The process of laser metal welding is widely used in industry. Nevertheless, there is still a lack of complete process understanding and control. For analyzing the process we used two high-speed cameras. Therefore, we could image the plasma plume (which is directly accessible by a camera) and the keyhole (where most of the process instabilities occur) during laser welding isochronously. Applying different image processing steps we were able to find a correlation between those two process characteristics. Additionally we imaged the plasma plume from two directions and were able to calculate a volume with respect to the vaporized material the plasma plume carries. Due to these correlations we are able to conclude the keyhole stability from imaging the plasma plume and vice versa. We used the found correlation between the keyhole behavior and the plasma plume to explain the effect of changing laser power and feed rate on the keyhole geometry. Furthermore, we tried to outline the phenomena which have the biggest effect on the keyhole geometry during changes of feed rate and laser power.

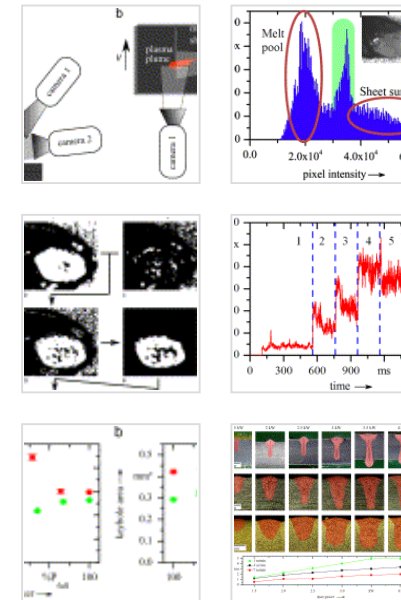
Keywords

Laser welding; High-speed camera; Image analysis; Keyhole; Plasma plume

References

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1. Introduction

Laser welding offers great flexibility and a high degree of automation. The process is highly dynamic and consists of complex multidimensional mechanisms which can lead to weld defects. Furthermore, the time span in which defects evolve (likely to be less than 1 ms) makes it hard to fully control the process and anticipate the formation of joint defects.

Therefore, complete process knowledge has to be developed with the use of high-speed imaging technologies. For reaching this goal not only a single part of the laser interaction zone has to be studied, but also the whole system of the emerging keyhole, the pool of molten metal around the keyhole and the plasma plume. The high process dynamics, however, makes it very difficult to image these characteristic phenomena. Observing keyhole fluctuations during laser welding is even more complicated due to the chaotic dynamics inside the keyhole and melt pool and the small diameter of the keyhole.

For applying the gained process knowledge in industry it is necessary to find a process phenomenon which is easy to measure *and* a good representation of the process behavior. Therefore, we imaged the plasma plume (which is directly accessible by a camera) and the keyhole (where most of the process instabilities occur) during laser welding isochronously. With this data we are able to build a model of the process and to give a statement about the link between these two process phenomena. As an ultimate goal we want to be able to extract the factors for keyhole stability just from sensing the plasma plume's behavior.

2. State of the art

The simulation and experimental analysis [1], [2] of laser metal welding is continuously improving and helps to increase the process understanding and therefore the process robustness in an industrial application.

New high-speed imaging systems made it possible to get an impression of the activities in the keyhole [3]. Furthermore, the behavior of the plasma plume was investigated [4], [5]. Fabbro et al. published a study where correlations between the plasma plume and the melt pool of liquid metal are shown [6]. However, the melt pool is just an indicator for a process situation which has already evolved in the keyhole. It is therefore not possible to prevent process instabilities that occur on short time scales by detecting the melt pool characteristics. For this purpose, the keyhole itself is better suited. In recent time the exploration of laser metal welding with X-ray analysis has been deepened [7], [8] and a

first attempt of three dimensional modeling of a keyhole using experimental data has been made [9].

Due to the fact that the plasma plume is emerging from the keyhole we postulate that there is a correlation between these two welding characteristics. Still, the link between plasma plume behavior and the keyhole stability has not yet been determined experimentally in a high spatial and temporal resolution.

3. Experimental

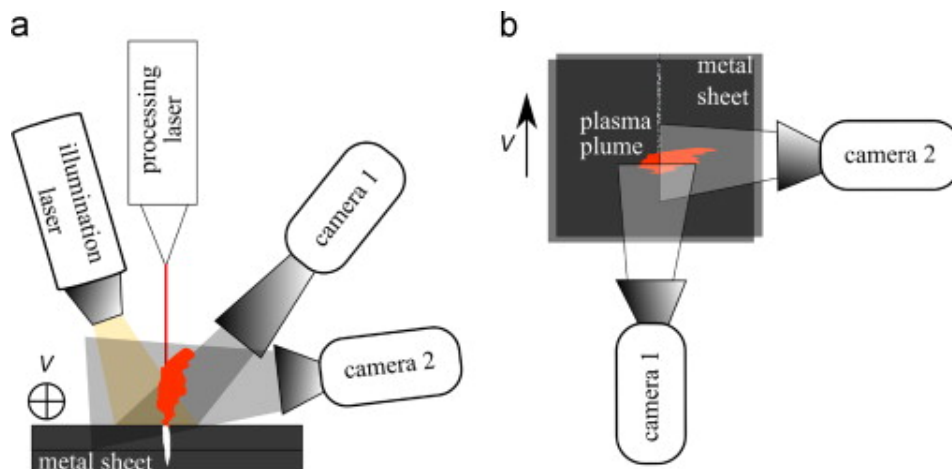
To study the dynamic behavior as well as the correlations between the keyhole and the plasma plume we used two high-speed cameras (VisionResearch Phantom v1210) working with a frame rate of 100kHz when using an image size of 256×256pixel. Both cameras are triggered at the same time to image the plasma plume and the behavior of the keyhole and the surrounding melt pool isochronously. We used a laser system (Cavitar Cavilux HF, 807nm, 500W peak power) to illuminate the keyhole region. Using a band-pass filter in front of the camera that imaged the keyhole region, we ensured that only the wavelength of the illumination laser reached the imaging chip of that camera. All other disturbing light (e.g. the brightness of the plasma plume and the reflections of the processing laser from the melt pool) were blocked or reduced.

For material processing we used a TruDisk 4002 Yb:YAG laser (wavelength 1030nm, maximum output power 4kW) coupled to focusing optics (BEO D70, focal length 200mm, spot size **600 μm** , beam profile *top hat*). We applied this setup to weld two stainless steel plates (X6CrNiTi18-10) of 1mm thickness in overlap configuration. Furthermore, we used an industry robot (Kuka KR125) for moving the metal sheets with constant feed rates of 1, 4 and 7m/min and increased the laser power step-wise to ensure the change from partial to full penetration in the last laser power step.

3.1. Imaging setups

Since the keyhole diameter is in the range of the laser spot size (**600 μm**) a magnification objective (factor 1.5) from Navitar was used to obtain a detailed view using a working distance of approximately 100mm. For imaging the plasma plume, which is compared to the keyhole much larger, we used a 100mm macro objective (Tokina ATX 2.8/100) in front of which we mounted a short pass filter to block reflections of the processing laser.

For imaging the keyhole and the plasma plume behavior two different camera setups were used. For the isochronous observation of the keyhole and the plasma plume camera 1 was lined up at a 50° angle to the metal sheets and the Navitar objective with its magnification of 1.5 was installed to have a detailed view of the keyhole. The optical axis of camera 2 was positioned at a 10° angle to see the plasma plume from its point of formation (the keyhole) to the top (see Fig. 1(a)). For measuring the plasma plume behavior in three dimensions we imaged the plume from the front and the side with the same type of camera and objective isochronously (see Fig. 1(a)).



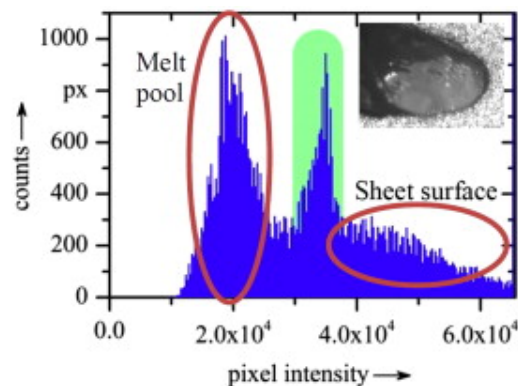
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Fig. 1. Imaging setups for isochronous keyhole and plasma plume observation (a) and the three dimensional plasma plume observation (b).

3.2. Image analysis

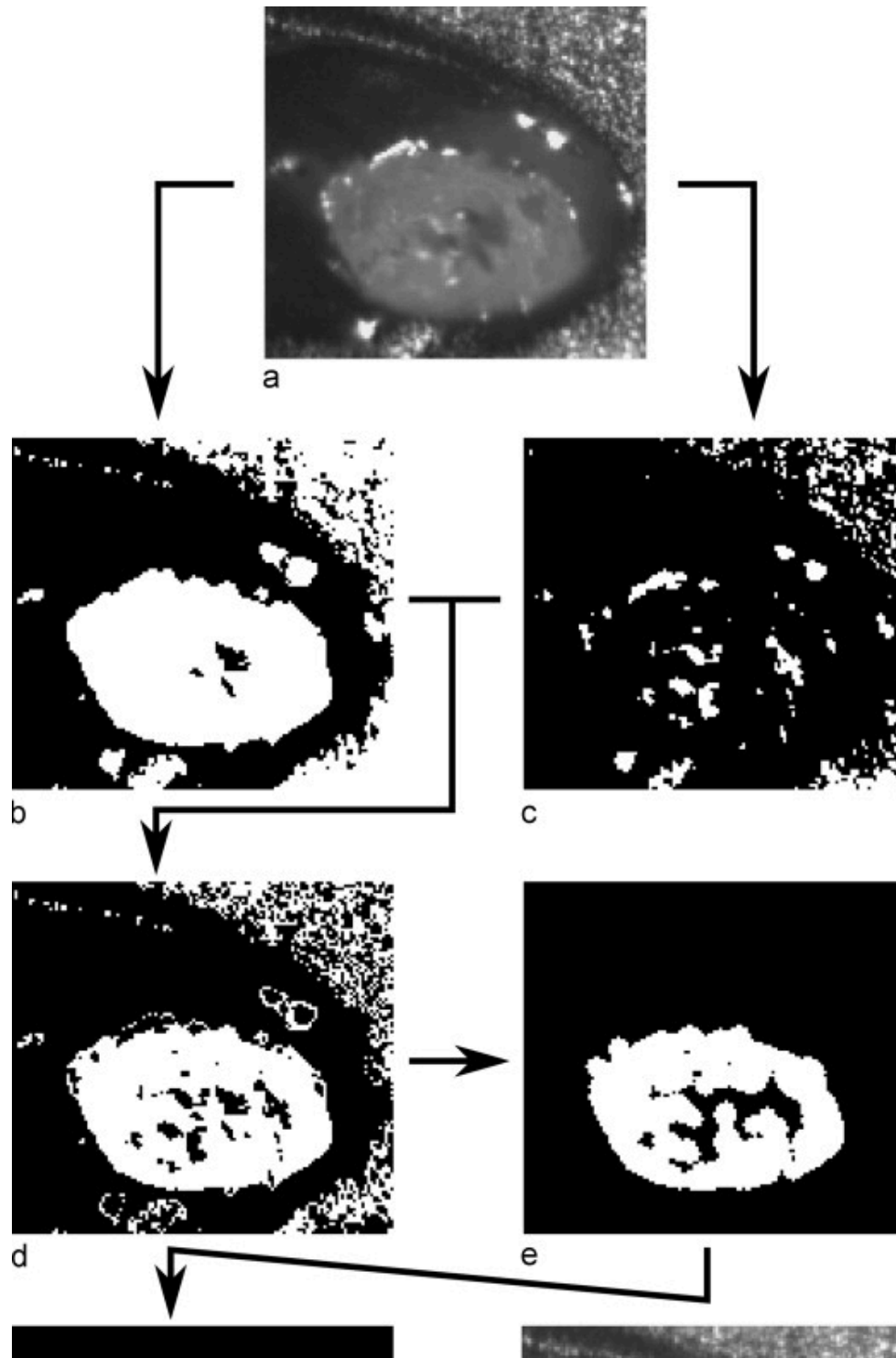
For evaluating the data we considered the area and the radiance of the plasma plume and the visible area of the keyhole. To analyze the dynamic behavior of the geometrical properties the scenery was imaged at high speed, so the uncertainty due to the high process dynamics was minimized (as compared with previous studies). Due to the high camera frame rate an algorithm was required to evaluate the 100000 images per second in a fast and reliable way.

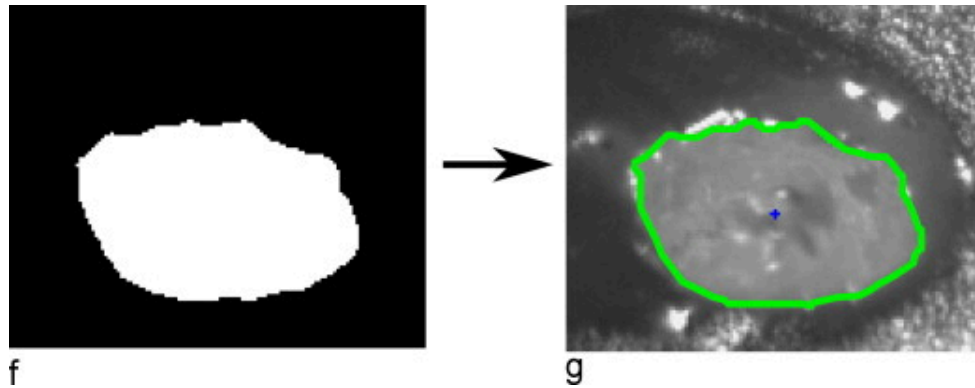
For evaluating the keyhole area we used an algorithm consisting of threshold operations and morphological filters [10] with Matlab. Fig. 2 shows a histogram of a typical keyhole image (higher pixel intensity means brighter gray values). Clearly visible are two peaks in the area of dark and medium gray values. The left peak originates from the dark area of the melt pool. This area is not directly illuminated by the processing laser and therefore much darker than the keyhole. The histogram counts to the right of the second peak derive from the speckled reflections of the metal sheet surface. However, the central peak of the histogram results from the bright keyhole region where the laser beam is coupled into the work piece and strong reflections occur. This peak could be obtained automatically to accelerate the image analysis. Fig. 3 shows the whole analysis process of a typical keyhole image.



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Fig.2. Histogram of a typical keyhole image (the higher the pixel intensity the brighter the image), clearly visible is the peak originating from the light gray keyhole region.





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Fig.3. Analysis process of a keyhole image: (a) original image, (b) pixels darker than the keyhole are removed, (c) pixels brighter than the keyhole are removed, (d) difference in (b) and (c), (e) eroded image, (f) dilated image, (g) recognized keyhole shape (green) with center of gravity (blue). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

In the first step we remove the left (pixels darker than the keyhole) and the right (pixels lighter than the keyhole) part of the histogram, shown in Fig. 3(b) and (c). By subtracting one from the other image we get the image shown in Fig. 3(d). Consequently, the black areas in the keyhole result from very light and dark pixel values. Nevertheless, the speckled sheet surface is still visible, due to the fact that some reflections have the intensity of the keyhole area. For automatic analysis of the keyhole area we need a closed white area surrounded by a constant black area. Therefore we used methods of morphological filtering. Fig. 3(e) shows the result of the so-called erosion. When eroding an image the black parts of the image grow and engulf small white areas (well visible in the metal sheet area). For obtaining a closed white area of the keyhole we have to dilate the image at the end. By dilating, the white parts of an image grow and engulf small dark areas (as shown in Fig. 3(f)). Fig. 3(g) reveals the final result. The keyhole area is very well recognized and characteristics like area, orientation, eccentricity and the center of gravity could be obtained.

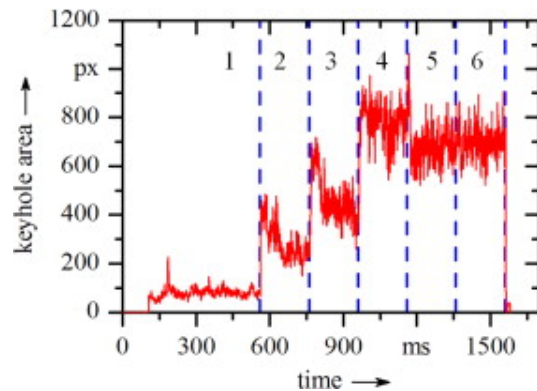
4. Results

For evaluating different welding setups we took a closer look at the visible area of the keyhole and the plasma plume.

4.1. Keyhole

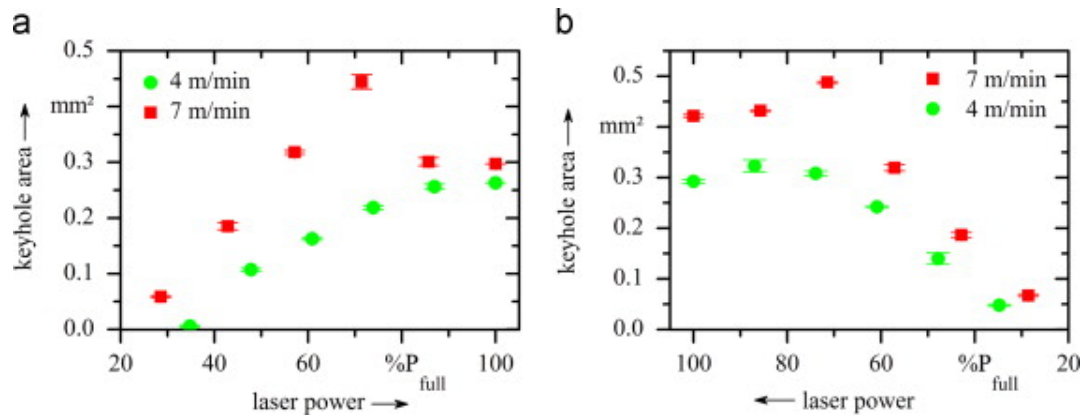
Since the keyhole is the area with the highest dynamics, and since here the fusion between the two metal sheets is initiated, we took a closer look at this area (see [Fig. 1\(a\)](#)). With the image analysis algorithm described before we were able to obtain the shape of the top of the keyhole. As already mentioned we did welds with a constant feed rate and ramped the laser power in a way that we reached full penetration of both sheets in the last power step. Due to the imaging frame rate of 100kHz and a step width of 200ms of the laser power, the characteristic process behavior within every power step could be analyzed with 20000 images. Despite the fact that we did not have a stable keyhole for every point of time during the weld, we did not filter any images for the studies we show in this paper. [Fig. 4](#) shows the raw data of the keyhole area detection for a feed rate of 7m/min and a stepwise increase of the laser power from 1 to 3.5kW in 500W steps. The changes of the keyhole area with rising laser power are clearly visible. In the first power step (till 580ms) a constantly open keyhole did not evolve and the algorithm did only detect the reflection of the illumination laser on the liquid top surface with a small diameter. In the beginning of step two and three, approximately the first 20ms, one can see that the keyhole area widens excessively. This behavior could not be seen in the high power steps. This led us to the conclusion that the initial excessive opening of the keyhole in the power steps two and three depends on the changes in the process, which are very strong in the beginning, when the keyhole opening area nearly doubles. Hence the keyhole does not extend that high in the power steps four to six the excessive opening cannot be seen. Furthermore, the keyhole area has a maximum when using 2.5kW and a subsequent decrease with increasing laser power. A detailed explanation of this behavior will be given in [Section 4.5](#). Not only the area, but also the fluctuation of the keyhole area changes with rising laser power. This is depending on a couple of reasons. Firstly, the

amount of molten material around the keyhole grows with rising power and the evaporation rate increases. Hence, the equilibrium between the pressure of the molten pool and the vaporization pressure inside the keyhole is more sensitive and the keyhole opening fluctuates more. Secondly, the keyhole deepens which leads to an increasing influence of the molten pool dynamics and an even more sensitive equilibrium and higher fluctuations. In Fig. 5(a) we compare the keyhole area for different feed rates. Every data point represents the mean value of 20000 evaluated images. To reach the same process result for the two different feed rates, the laser power was increased from 0.8 to 2.3kW in steps of 300W maintaining a feed rate of 4m/min. The laser power steps for a feed rate of 7m/min were the same as used before.



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Fig.4. Evolution of the keyhole area when ramping the laser power stepwise from 1 to 3.5kW with a constant feed rate of 7m/min on two 1 mm stainless steel plates (the red lines show the power steps). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)



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Fig.5. Visible keyhole area for 4 and 7 m/min feed rates on two 1 mm stainless steel plates, the laser power is given in percentage of the power needed for full penetration for the respective feed rate, power was increased (a) or decreased (b) during the weld, $n=3$.

The behavior of the keyhole area for a feed rate of 7 m/min is the same as in Fig. 4. For 4 m/min the keyhole area increases less and more constantly. Both experiments have in common that the increase of the keyhole area stops when reaching a threshold laser power P_{thresh} (approx. 80% of the power required for full penetration). We used this threshold power as a criterion for our following experiments and not the laser intensity, since we did not yet know the specific size of the illuminated area on the keyhole front wall.

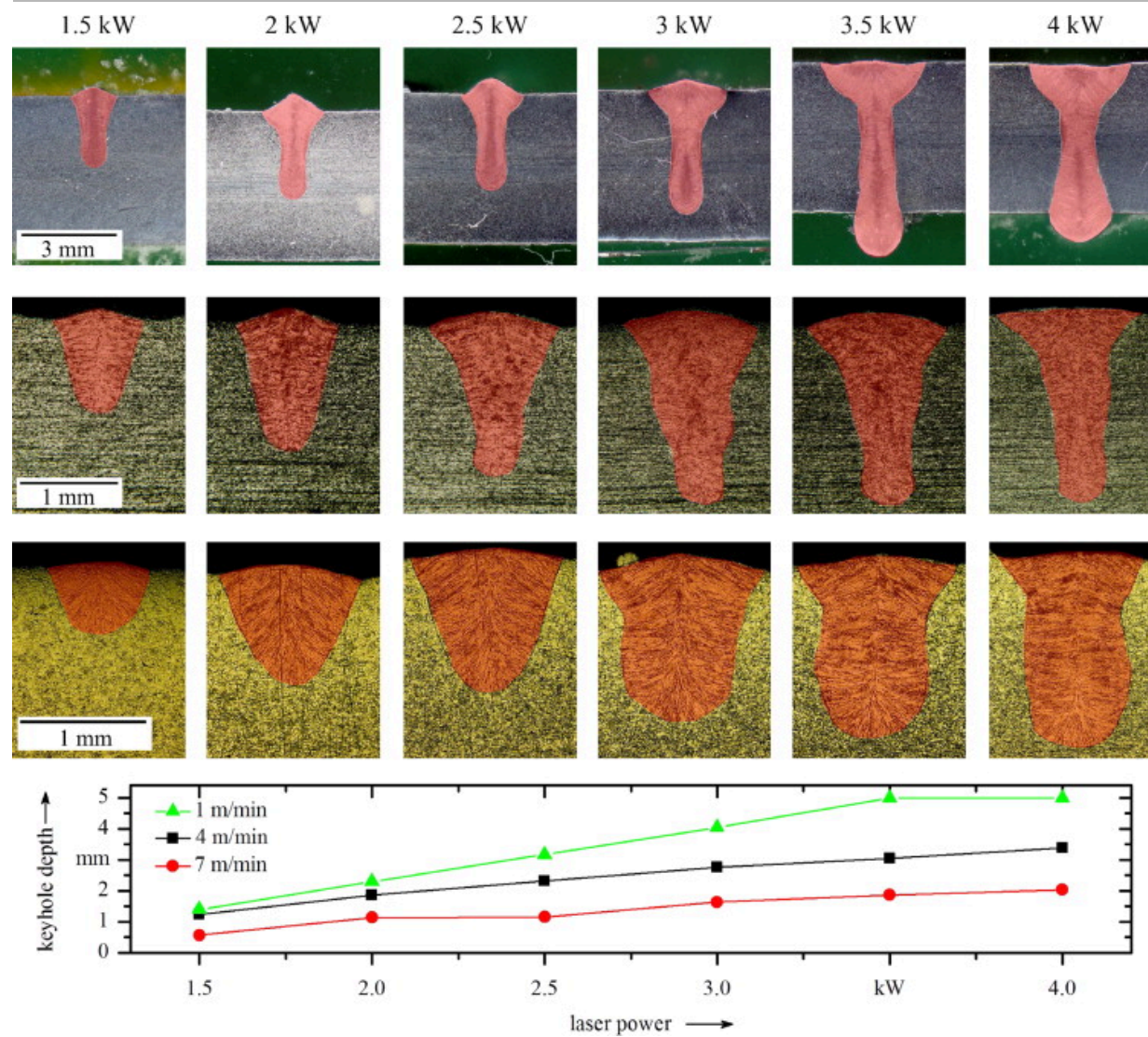
The experiments shown in Fig. 5(a) were also carried out with a stepwise *decrease* from highest to lowest laser power. Fig. 5(b) shows the results. The trend of the keyhole area evolution is the same as shown in Fig. 5(a). The visible keyhole opening area is nearly constant for the two highest laser power steps. The change in the trend of the keyhole area is independent of whether the laser power is increased or decreased. For a feed rate of 4 m/min the area decreases when reaching approximately 85% of P_{full} . For a feed rate of 7 m/min the turnaround starts when reaching 70% of P_{full} . Nevertheless, the percentage

increase of keyhole opening area from the two highest laser power steps to the third highest laser power step is smaller for 7m/min when decreasing the laser power (32% for increasing and 13% for decreasing laser power). This may arise from a different changing behavior of the keyhole geometry when decreasing the laser power. A decreasing laser power leads of course to a decrease in melt pool size. This leads to different starting conditions for the third highest laser power step (small melt pool for increasing laser power, large melt pool for decreasing laser power) which affects the evolution of the keyhole opening (smaller increase due to the higher pressure of the surrounding melt pool). The slightly different numerical values for the keyhole area in comparison to Fig. 5(a) originate from a slightly different alignment of the cameras (steeper inclination angle in Fig. 5(b)) and therefore an increase of the visible keyhole opening area. The difference of the keyhole area in the lowest laser power step (for example 0.04 mm^2 for a feed rate of 4m/min for decreasing laser power in comparison to nearly zero for increasing laser power) derives from the piercing operation. When starting with a high laser power the piercing operation can be done very easily and the keyhole is kept open also for a comparably low laser power. In the other case of starting with a low laser power a stable keyhole does not evolve due to reflections and the lack of the keyhole which leads to an increasing absorption.

Preheating effects that may affect the keyhole opening area can be neglected in our case for two reasons. Firstly, the temperature inside the keyhole is four to five times higher than the temperature in the surrounding metal sheet (especially in the area in front of the keyhole) [11]. Secondly, by calculating the mean value of the keyhole opening area for whole laser power step we minimize the initial effects for an increase in laser power.

For evaluating the influence of full penetration (where a part of the vaporized material can escape through the bottom of the keyhole) we performed a second set of experiments. For these experiments we used one 5mm thick stainless steel plate and ramped the laser power identically for every feed rate (from 1.5 to 4kW in 500W steps). Fig. 6 shows the resulting seam cross-sections and the seam depths for three different feed rates. Full penetration was reached solely for 3.5 and 4kW when using a feed rate of

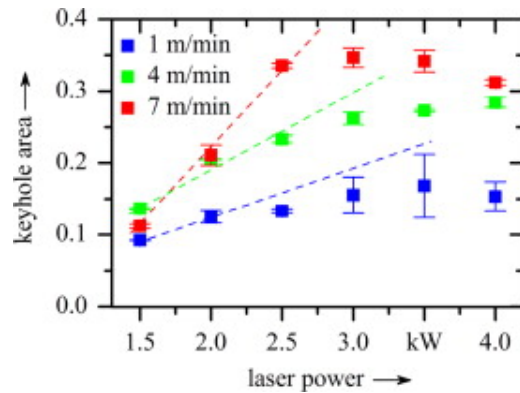
1 m/min. It is visible that the aspect ratio (seam depth/seam width) is increasing with increasing laser power and especially with a lower feed rate. Since the keyhole's cross-section is a measure how well the beam couples into the material these results show that the changing trend of the keyhole opening area when reaching P_{thresh} cannot be explained with a change in the beam-metal interaction.



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Fig.6. Cross-sections of the weld seams with 1 (upper row), 4 (middle row) and 7m/min (lower row) feed rates on one 5mm stainless steel plate with increasing laser power (weld seam cross-section colored); lower graph shows the seam depth with increasing laser power for 1, 4 and 7m/min. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

Fig. 7 shows the resulting keyhole area of these welds. The keyhole opening area obtained for feed rates of 4 and 7m/min is consistent with the data obtained when welding two 1mm sheets in overlap configuration. Furthermore, a decreasing keyhole opening area for a feed rate of 1 m/min is not visible before reaching 4kW (whereas the full penetration starts at 3.5kW). We therefore conclude that the transition to full penetration does not significantly affect the visible keyhole opening area. Since the decreasing trend in the keyhole opening area is starting before the full penetration of the two metal sheets we believed that this decrease is not solely depending on the penetration status of the weld. The detected keyhole area is approximately the same in the first power step for all three feed rates. When increasing the laser power at the lowest feed rate, the visible keyhole area increases only slightly. The welds performed with feed rates of 1 m/min and 4m/min show saturation in the keyhole area when reaching 3kW. The weld with 7m/min even shows a decrease of the keyhole area when exceeding the threshold laser power P_{thresh} . The fluctuation of the keyhole area, quantified using the standard deviation, is higher for 1m/min than for 4m/min due to the fact that the molten pool around the keyhole for low feed rates is very large (approximately three times the diameter of the keyhole). This mass of liquid metal has strong dynamics and changes the area of the open keyhole significantly.



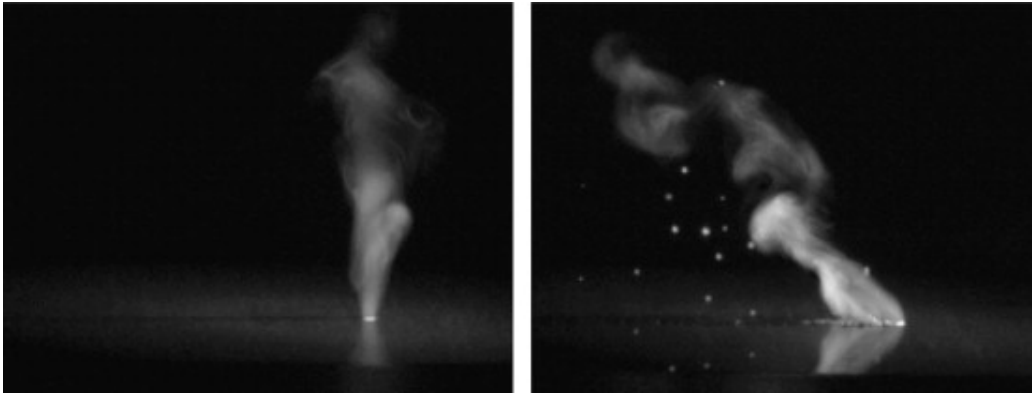
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Fig.7. Visible keyhole area for 1, 4 and 7 m/min feed rates on one 5 mm stainless steel plate. The dashed lines show the trend of the keyhole opening area for the first laser power steps, $n=3$.

4.2. Plasma plume

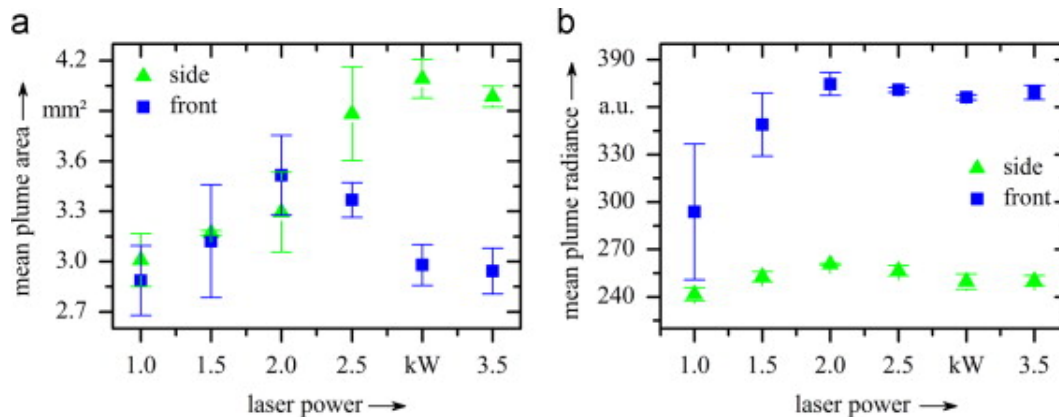
The plasma plume is the indicator for the state of the laser welding process which is easiest to image. This material stream out of the keyhole may be a gauge for the amount of material vaporized in the keyhole. Therefore we imaged the plume from the front and the side with the same type of camera and objective isochronously (see Fig. 1(a)). For evaluating the plume area A_{plume} we used an algorithm related to the one for keyhole area determination. The area of the plasma plume imaged by one camera alone is only usable for a rough estimation of the vaporized material. Fig. 8 shows the behavior of the plume when increasing the feed rate. As already shown in [5] the inclination changes with increasing laser power. Therefore, the plume area which is visible from the front A_{front} decreases (see Fig. 9(a)). When using the mean plume areas A_{side} and A_{front} for a volume estimation alone it would contradict the fact that the amount of vaporized material increases with increasing laser power [12], [13]. Therefore, we have to take into account the plume radiance L_{plume} as another crucial factor. Furthermore, this factor can be used to

meet the fact that the plume may contain the same mass when it is large and thin or small and dense (see Fig. 9(b)).



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Fig.8. Images recorded by camera 2 from the side. The plume inclination and the radiance change with rising laser power: laser power 1.5 (left) and 3.5kW (right).



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Fig.9. Mean plume area (a) and radiance (b) for different laser powers and a constant feed rate of 7m/min on two 1 mm stainless steel plates, imaged by camera 1 (front) and camera 2 (side), $n=3$.

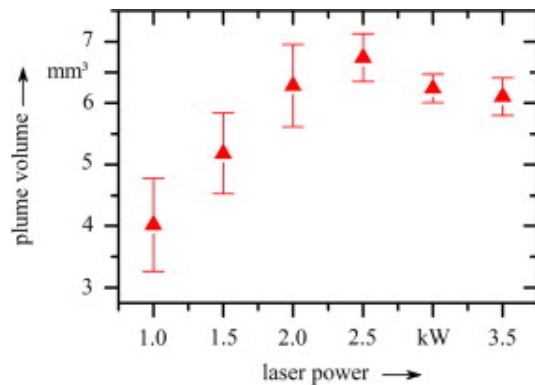
When combining these two indicators we get for both the cameras

$$A_{v,side} = A_{side} \frac{L_{plume,side}}{\max(L_{plume,side})} A_{v,front} = A_{front} \frac{L_{plume,front}}{\max(L_{plume,front})}$$

Consequently, we can calculate the estimated virtual volume of the plasma plume:

$$V_{plume} = \sqrt{A_{v,side}} \sqrt{A_{v,front}} \sqrt{\frac{A_{v,side} + A_{v,front}}{2}}$$

Fig. 10 shows the estimated virtual volume for the weld shown in Fig. 4. The volume of the plasma plume increases until the threshold laser power of 2.5kW and decreases after this power step in the same way as the keyhole area does (see Fig. 5). This shows that the abnormality of the keyhole opening area is not a measuring failure but the real process behavior.

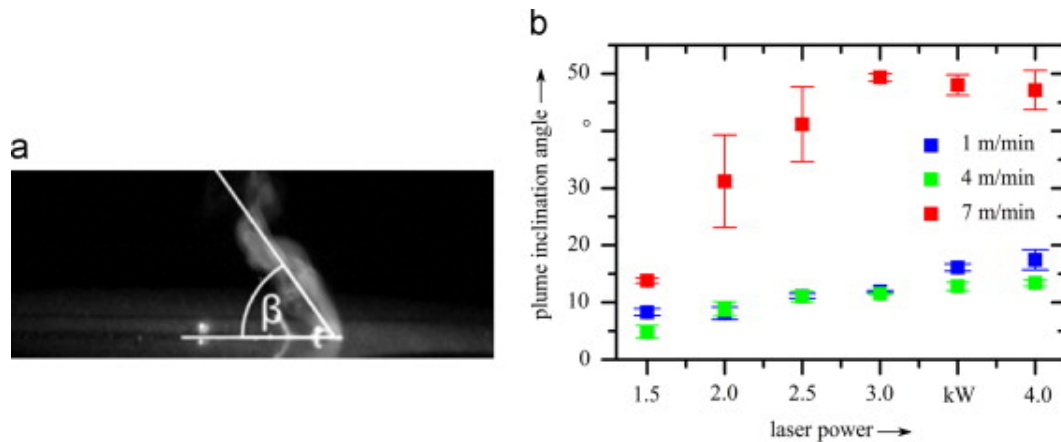


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Fig. 10. Calculated plume estimated virtual volume for different laser powers and a constant feed rate of 7m/min on two 1 mm stainless steel plates, $n=3$.

For a deeper analysis we characterize the plume not only by its radiance and volume, but also by its inclination in opposite welding direction β (as indicated in Fig. 11(a)). A first investigation of this inclination was already published in [4], [5]. In the current experiments the inclination of the plume was also observed for welds with 5 mm thick

plates with the same parameters as shown in Fig. 7 (feed rates 1, 4 and 7 m/min and stepwise increase of laser power from 1.5 to 4 kW in 500 W steps). Fig. 11(a) shows the inclination angle for these power steps. It is obvious that the plume's inclination is depending on the laser power and the feed rate. With low feed rates of 1 and 4 m/min the inclination changes just slightly with increasing power. When using a higher feed rate of 7 m/min a much greater dependency on the laser power could be observed (as in case of the keyhole opening area). Furthermore, it could be seen that the trend of the inclination changes when reaching a threshold laser power of 3 kW.



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Fig. 11. (a) Plume inclination angle β (imaged by camera 2); (b) plume inclination angle β for 1, 4 and 7 m/min feed rates on one 5 mm stainless steel plate, $n=3$.

4.3. Correlation between keyhole and plasma plume

In the previous section we showed that the behavior of the keyhole and the plasma plume is of similar type when changing laser power and feed rate. For a prediction of the keyhole behavior from the sensing of the plasma plume a correlation with a high temporal resolution is needed. Therefore, we cross-correlated [14] the signals from the

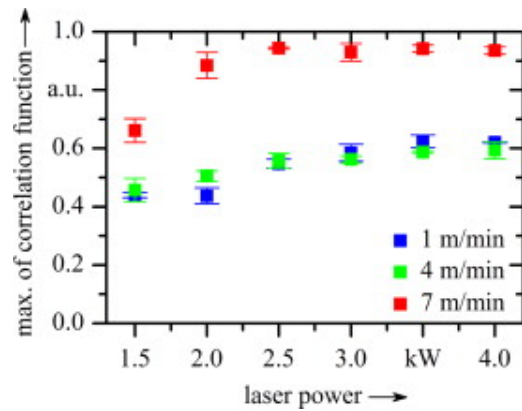
keyhole and the plasma plume which we measured with 100kHz:

$$\hat{R}_{xy}(k) = \sum_{n=0}^{N-1-k} y_{n+k} x_n$$

where N is the length of the measured signals (20000 samples per step of the laser power ramp in our case) and the two signals x and y are keyhole opening area and the plasma plume inclination angle β from the experimental data shown in [Fig. 7](#) (feed rates 1, 4 and 7m/min and stepwise increase of laser power from 1.5 to 4kW in 500W steps).

R_{xy} was normalized in a way that the auto-correlations of the individual signals at zero lag ($k=0$) are identically one. Due to the high frame rate of our [imaging system](#) we could measure the keyhole area and the plume inclination in a very continuous way so that no steps in the signals occur and a steady cross-correlation function could be calculated. One has to keep in mind that a high value of the cross-correlation function is just an indicator for the correlation between the keyhole opening area and the plasma plume inclination and not for a stable welding process.

[Fig. 12](#) shows the maximum of the normalized cross-correlation function for different feed rate and laser power settings. In each case, the maximum cross-correlation was obtained for $k=0$. The correlation between the keyhole opening area and the plume inclination is increasing with increasing laser power as well as with increasing feed rate. The analysis shows that the two phenomena are well correlated when using a feed rate of 7m/min, especially for high values of laser power. In case of lower feed rates the keyhole opening is mainly affected not only by the plasma plume, but also by the other phenomena like melt pool dynamics and the flow of molten material around the keyhole [\[15\]](#).



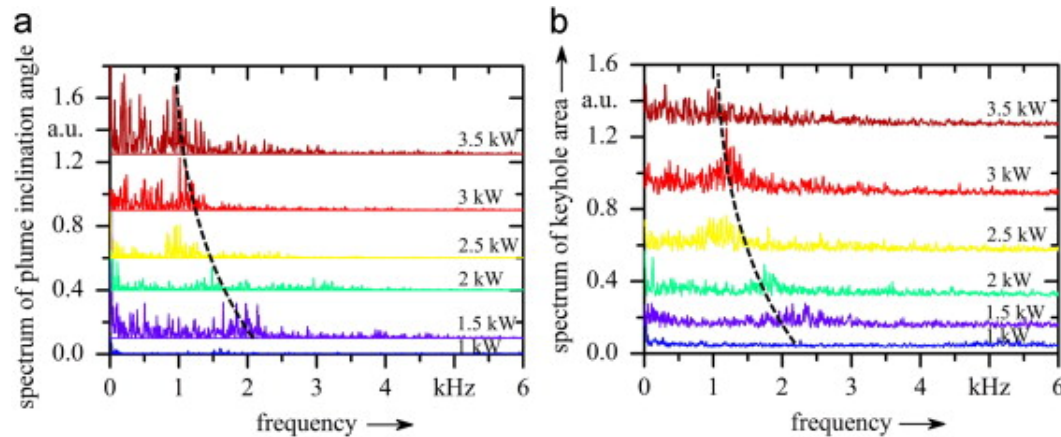
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Fig. 12. Maximum of the cross-correlation function R_{xy} between the keyhole opening area and the plume inclination angle β for 1, 4 and 7 m/min feed rate on one 5 mm stainless steel plate, $n=3$.

4.4. Dynamic behavior

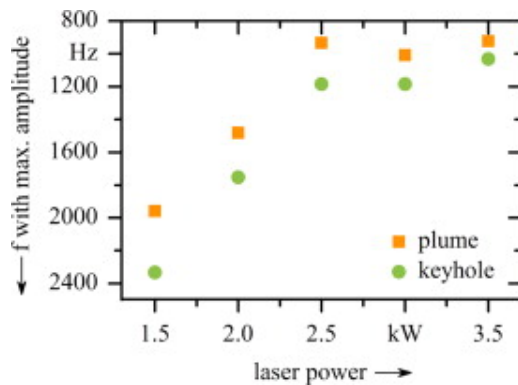
When using high-speed cameras the dynamic behavior of the process is a very interesting topic. Therefore we compared the plume inclination angle β and the keyhole area by applying frequency analysis, using the data from the experiment with the 5 mm steel sheets (Fig. 7). To quantify the dynamic behavior we used Discrete Fourier Transform computed with a Fast Fourier Transform (FFT) algorithm in **MATLAB**[®]. The resulting single-sided amplitude spectrum was analyzed for every power step. From other publications we know that the keyhole is oscillation with frequencies higher than 500 Hz [16]. Therefore, we defined the characteristic frequencies for every power step as the maximum frequency in the range of 0.5–3 kHz. Fig. 13 shows the FFT for a weld with a constant feed rate of 7 m/min and a stepwise rise of laser power from 1 kW to 3.5 kW. The data obtained during every single power step was analyzed separately. The maximum of the frequency spectra for every power step in the range from 0.5 to 3 kHz are shown in Fig. 14. One can clearly see that the trend of the frequency spectra changes when reaching P_{thresh} . Two conclusions can be drawn from this analysis. On the one hand the dominating

frequencies in the plume inclination spectra and in the keyhole area spectra follow the same trend. On the other hand one can clearly see that there are no characteristic frequencies above 4kHz. Even when we look at the whole obtainable spectrum up to 50kHz there could not be any frequency peak found related to the process.



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Fig.13. Spectra of the plume inclination angle (a) and the visible keyhole area (b) for a constant feed rate of 7m/min on one 5mm stainless steel plate. The spectra are plotted to separate them best, the dashed line indicates the trend of the dominating frequencies (no dominating frequency visible for 1 kW).



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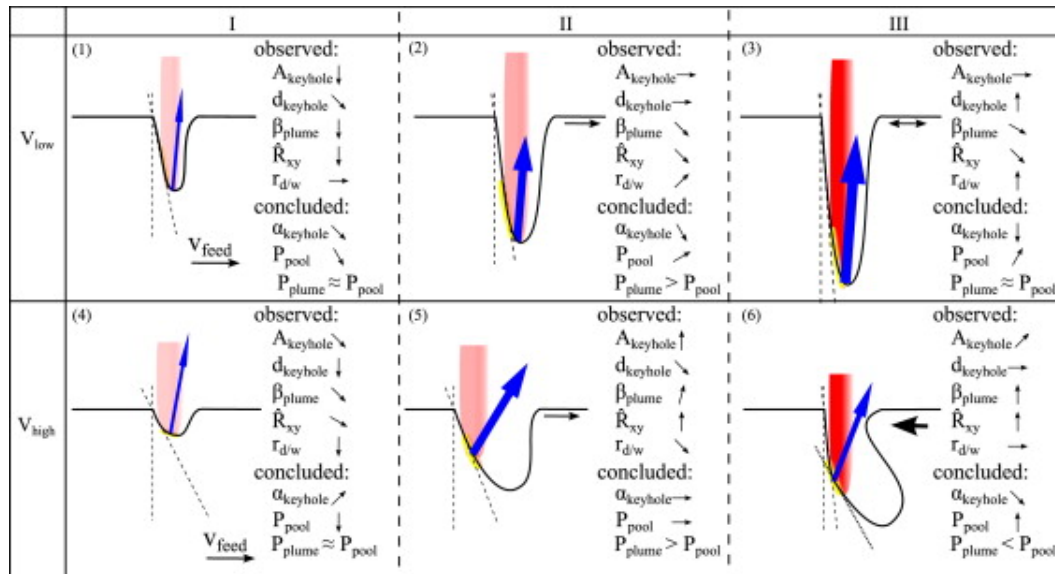
Fig. 14. Maxima of the spectra of the plume inclination angle and the keyhole area in dependency of the laser power (y-axis reversed for better comparison with Fig. 5(a)).

Consequently, our analysis shows that the dynamic behavior of the keyhole and the plume is well correlated, especially for high laser powers and feed rates. Additionally, it could be measured that the trend of both, the keyhole and the plasma plume behavior, changes when reaching a threshold laser power P_{thresh} .

4.5. Modeling of the keyhole geometry

After the observation and analysis of the weld seam width and depth, the keyhole opening area, the plume inclination angle and the frequency analysis and cross-correlation between those two values and the estimation of the plasma plume volume, we use the results to model the keyhole geometry. Furthermore, we try to outline the phenomena which have the biggest effect on the keyhole geometry during changes of feed rate and laser power.

Fig. 15 shows the conclusion of our modeling, where v_{low} is a summary of the feed rates 1 and 4 m/min and v_{high} is the representation of 7 m/min for the experiments with one 5 mm thick steel sheet. The increase in laser power is separated into three regimes, where each regime includes two laser power steps. For each domain the geometry of the keyhole, the processing laser and a representation of the main orientation of the plasma plume (blue arrow) are shown. It must be mentioned that this is an idealized representation due to the wavy keyhole front [17] which leads to a more complex behavior in reality. In every regime we illustrate the trend of the keyhole opening area ($A_{keyhole}$), the depth of the keyhole depicted in the cross-sections ($d_{keyhole}$), the plume inclination angle (β_{plume}), the maximum of the cross-correlation function (R_{xy}) and the calculated aspect ratio ($r_{d/w} = d_{keyhole}/w_{seam}$), where w_{seam} is the seam width in the cross-sections (Fig. 6).



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Fig. 15. Evolution of the keyhole geometry in the case of constant feed rate and increasing laser power (model based on the experimental results). The red bar indicates the processing laser and the blue arrow represents the resulting plasma plume, v_{low} is equal to feed rates 1 and 4 m/min and v_{high} is equal to 7 m/min. The laser power increases with increasing regime number (I–III), each regime includes two laser power steps. In every regime the trend of the keyhole opening area ($A_{keyhole}$), the depth of the keyhole obtained from the cross-sections ($d_{keyhole}$), the plume inclination angle (β_{plume}), the maximum of the cross-correlation function (R_{xy}) and the calculated aspect ratio ($r_{d/w} = d_{keyhole}/w_{seam}$) is shown. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

Besides the measured values we assessed the inclination of the keyhole front wall ($\alpha_{keyhole}$). The inclination is influenced by heat transfer from the point of incidence of the processing laser to the bottom of the keyhole. When increasing the laser power or decreasing the feed rate, more heat is transferred to the keyhole bottom. This rise of

energy helps to melt more material at the bottom and erects the keyhole, as already stated in [6], [18].

The saturation of the keyhole area growth for v_{low} and the shrinking for v_{high} is clearly visible when reaching a threshold laser power P_{thresh} (regime III). Assuming that the estimated plasma plume volume V_{plume} is linearly proportional to the evaporation rate, we can understand the evolution of the visible keyhole area shown in Fig.4, Fig.5, Fig.7 in accordance with V_{plume} in Fig. 10. As the evaporation rate grows, so does the recoil pressure inside the keyhole [19]. Consequently, the keyhole expands as the evaporation rate increases. Hence, the keyhole opening should grow with increasing laser power when using a constant feed rate. Nevertheless, we experienced a saturation of the visible keyhole area when reaching a threshold laser power. Therefore, there must be a change of the factors which determine the keyhole shape during our weld experiment.

The recoil pressure is the most important factor for keeping the keyhole open [19] whereas the pressure of the molten material around the keyhole acts as a counter-force. Only if these pressures are balanced a stable keyhole can evolve. Nevertheless, there must be an initial change of this equilibrium when process parameters and thus the keyhole opening area change. Therefore, the keyhole opening area is an indicator of the relationship between the recoil pressure P_{plume} and the counter-pressure P_{pool} . For $P_{plume} > P_{pool}$ the keyhole expands till the pressures are balanced whereas for $P_{plume} < P_{pool}$ the keyhole opening area is shrinking or the keyhole is even collapsing. All these presumptions lead to the following model:

v_{low} I: The keyhole is just slightly inclined and has a medium aspect ratio (depth/diameter). Therefore, the plume is mostly directed vertically and does not affect the melt pool (and thus the keyhole opening) very much so the keyhole opening is small and no strong correlation can be observed.

v_{low} II: When increasing the laser power the keyhole is still just weakly inclined whereas the keyhole depth increases with just a small growth in diameter (increase in aspect ratio). Since the laser power is increased the evaporation pressure inside the keyhole

increases as well. Due to the narrow keyhole the evaporation pressure does affect the molten pool stronger and the correlation as well as the keyhole opening area increases.

v_{low} **III**: A further increase of the laser power leads to an even deeper keyhole. Condensation (as shown in [12]) and loss of momentum of the vapor particles may then lead to a decreasing evaporation rate. Due to these losses the resulting evaporation pressure stays constant and the keyhole opening area does not increase further.

v_{high} **I**: For a high feed rate the keyhole is more inclined than for the before mentioned case. Due to this inclination of the keyhole the plume is inclined as well and is more directed to the keyhole rear wall. Thus, the keyhole opening area is more correlated with the plasma plume.

v_{high} **II**: When increasing the laser power the keyhole inclination gets weaker, the plume inclination angle rises and the plume gets even more directed into the keyhole rear wall. Therefore, the keyhole opening area extends and the correlation increases rapidly.

v_{high} **III**: A further increase in laser power leads to a further erection of the keyhole which should actually increase the inclination of the plasma plume and the keyhole opening area. The fact that we observed a contrary behavior of these phenomena led us to the conclusion that a threshold laser power is reached, where the upper half of the keyhole front wall slope decreases rapidly and the relations from [6], [18] are no longer valid anymore. This change in the keyhole geometry, from a front wall with a constant slope gradient to a step in the slope gradient, results in a change of the area which is directly illuminated by the processing laser. The point of incidence moves deeper into the keyhole to a position where the keyhole inclination is comparable with the inclination in regime II. Therefore, the plume inclination does not change significantly between these two laser power regimes. This deep point of incidence leads to condensation inside the keyhole and thus to a loss of momentum of the evaporation pressure. Therefore, the counter-pressure against the surrounding molten pool gets weaker which consequently leads to a decrease of the keyhole opening area. Nevertheless, the correlation between the plume inclination and the keyhole opening area is still high, because the plume is still directed onto the

keyhole rear wall. However, the correlation is now dominated mostly by the pressure of the surrounding molten pool and not any more by the evaporation pressure. This shows that a high correlation is evidence for a specific process situation and not for a stable process in general. The fact that the keyhole area does not decrease for v_{low} **III** is caused by the much smaller $\alpha_{keyhole}$ for these low feed rates. A stepwise increase of the slope gradient with an effect on the keyhole behavior not possible is in that case because the keyhole is already nearly vertically directed.

5. Conclusion and outlook

We showed an experimental approach using two high-speed cameras to isochronously investigate the behavior of the plasma plume and the keyhole during laser welding. Using several image processing steps, we were able to extract the relevant information on the keyhole's visible top area.

A strong dependency of the size of this area on the applied feed rate and laser power could be revealed and expressed quantifiably. For a further process understanding we took a closer look at the plasma plume as an indicator for the amount of vaporized material ejected out of the keyhole. With simultaneous imaging from the front and the side we could successfully reconstruct an estimated virtual volume of the plume.

With cross-correlation analysis we found out that the dynamic behavior of the keyhole and the plume is well correlated, especially for high laser powers and feed rates. Additionally, it could be measured that the trend of both, the keyhole and the plasma plume behavior, changes when reaching a threshold laser power P_{thresh} .

The changing correlations between the plume and the keyhole with changing laser power and feed rate led us to the conclusion that there must be a change of the factors which determine the keyhole shape. We conclude that the keyhole stability for low and high feed rates with low to medium laser power is predominantly determined by evaporation processes in the keyhole. When using a high feed rate and a high laser power the

determining factor of the keyhole stability is the melt pool. Furthermore, our measurements indicate the occurrence of a step in the slope gradient of the keyhole front wall for these process parameters. Our study shows that the plasma plume is a good measure for determining the keyhole behavior, especially for feed rates where the recoil pressure dominates the geometrical shape of the keyhole. The ultimate goal to extract factors for keyhole stability just from sensing the plasma plume's behavior was not yet reached. Nevertheless, we were able to conclude the keyhole stability from imaging the plasma plume and vice versa.

For our future work we would like to take a deeper look on the material dependency of the measured values, especially the kinking of the keyhole front wall. Furthermore, a next step for defect prevention should be made by analyzing the kinking behavior with respect to the occurrence of defects. For a deeper understanding of the monitored behavior we would like to compare our results with welding simulations and observe the melt and gas dynamics inside the keyhole.

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